

PERSONAL PROJECT PORTFOLIO

AI-Driven Intrusion Detection System: Machine Learning Pipeline for Network Anomaly Detection

Toochukwu Praise Ajoku

MSc Cyber Security | Keele University
Student Member, CIISec
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Dataset	CIC-IDS2017 Friday Afternoon DDoS (225,741 flows)
Models	Random Forest (supervised) + Isolation Forest (unsupervised)
Explainability	SHAP TreeExplainer - global beeswarm + individual waterfall plots
Performance	RF: F1=0.9999, ROC-AUC=1.0000 IF: F1=0.375, AUC=0.2825
Key Skills	Feature engineering, class imbalance handling, adversarial robustness, XAI
Frameworks	MITRE ATT&CK, BCS Code of Conduct, UK GDPR / data minimisation

Executive Summary

This portfolio documents a self-directed machine learning project in which I designed, implemented, and critically evaluated an AI-driven Intrusion Detection System (IDS) pipeline against the CIC-IDS2017 Friday Afternoon DDoS dataset. The project directly addresses a priority gap in the UK cyber security job market: the ability to combine data science techniques with security domain expertise to build explainable, production-ready threat detection systems.

The pipeline was motivated by a realistic SOC scenario: Menzis Finance facing escalating network intrusions that evade traditional signature-based detection. I implemented two complementary detection strategies - a supervised Random Forest (RF) for high-precision real-time classification, and an unsupervised Isolation Forest (IF) for zero-day pattern detection where no labelled attack samples exist - directly addressing the four CISO requirements of precision, explainability, low false positives, and unseen-pattern detection.

SHAP TreeExplainer was applied to the RF model to produce both global feature importance rankings and per-prediction waterfall plots, providing the analyst-facing explainability layer that distinguishes a deployable SOC tool from an academic experiment. The pipeline achieved near-perfect supervised detection (RF F1=0.9999, ROC-AUC=1.0000) with only 3 misclassifications across 45,149 test samples (Lundberg and Lee, 2017).

The critical reflection addresses the methodological rigour decisions made throughout - including contamination leakage correction, the rejection of K-Means and XGBoost as alternatives, and the ethical dimensions of false positives as a denial-equivalent harm - positioning this project as evidence of both technical capability and security-informed engineering judgement.

Part 1: Dataset & Problem Definition

1.1 Dataset Overview

The CIC-IDS2017 dataset, produced by Sharafaldin, Lashkari and Ghorbani (2018) at the Canadian Institute for Cybersecurity, is the most widely used benchmark for network intrusion detection research. Its realistic traffic generation methodology - using actual benign background traffic rather than synthetic approximations - makes it more representative of production network conditions than earlier IDS benchmarks.

For this project I worked with the Friday Afternoon subset, which captures a DDoS attack scenario against a target network. The dataset contains 225,741 network flow records with 78 numeric features extracted from packet captures, plus a Label column. The class distribution presents a significant modelling challenge: 128,027 DDoS flows (56.71%) versus 97,714 BENIGN flows (43.29%) - a class imbalance that, if unaddressed, biases classifiers toward the majority class (Stallings and Brown, 2018).

Label	Count	Proportion
DDoS	128,027	56.71%
BENIGN	97,714 (4 rows dropped)	43.29%
TOTAL	225,741	100.00%

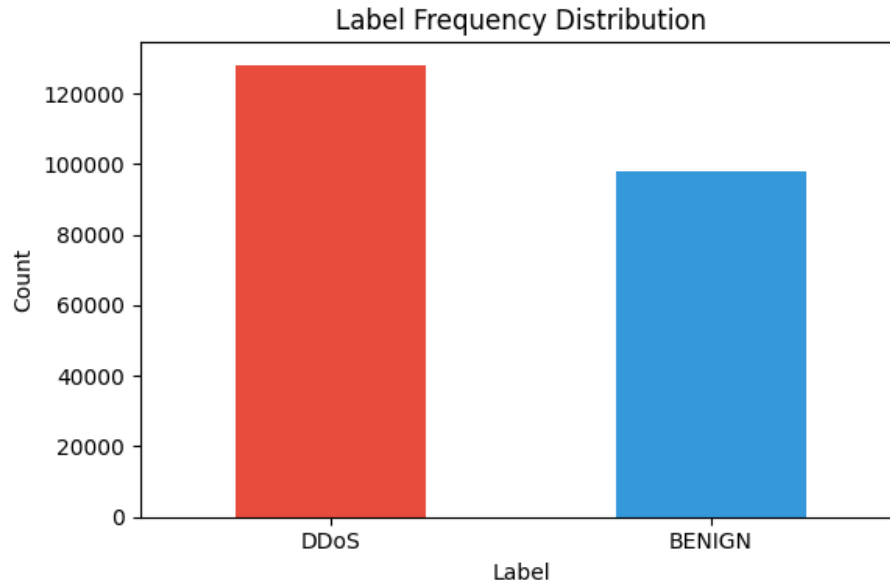


Figure 1: Label distribution - CIC-IDS2017 Friday Afternoon (225,741 flows). Class imbalance (56.7% DDoS / 43.3% BENIGN) confirmed, stratified split applied to preserve ratio

1.2 CISO Requirements & Detection Strategy

The project was framed around four CISO requirements that represent realistic production deployment constraints, rather than optimising for benchmark accuracy alone:

- Precision: minimise false positives that generate analyst alert fatigue and block legitimate traffic (Capuano et al., 2022).
- Explainability: provide per-alert reasoning that allows a Tier-1 SOC analyst to validate or dismiss alerts within sub-minute timeframes (Lundberg et al., 2020).
- Low false positives: false positives in a network IDS create denial-equivalent harm for legitimate users - an ethical consideration as well as a technical one (Stallings and Brown, 2018).
- Unseen-pattern detection: the pipeline must identify novel attack patterns for which no labelled training samples exist - addressing the fundamental limitation of supervised-only approaches (Liu, Ting and Zhou, 2012).

These requirements motivated the dual-model architecture: RF addresses requirements i–iii; IF addresses requirement iv. Neither model alone is sufficient.

Part 2: Data Preparation & Feature Engineering

2.1 Data Loading & Cleaning

The dataset was loaded from the CIC-IDS2017 Friday Afternoon CSV file. Four preprocessing steps were applied before any feature engineering:

1. Column name whitespace stripping - a known CIC-IDS2017 artefact where column names contain leading/trailing spaces that cause silent key mismatches in pandas (Sharafaldin, Lashkari and Ghorbani, 2018).
2. Missing value removal - 4 rows dropped (0.002% of dataset), negligible impact on class distribution.
3. Stratified 80/20 train-test split (180,592 train / 45,149 test) - preserving the 56.71%/43.29% DDoS/BENIGN ratio in both sets to avoid biased evaluation metrics (Tsukerman, 2019).
4. Test set isolation - no preprocessing parameter (scaler means, feature selector thresholds, contamination ratio) was derived from the test set at any stage, preventing data leakage (Stallings and Brown, 2018).

```
#Load the raw dataset from local storage
df = pd.read_csv(
    r"C:\Users\ajoku\OneDrive\Documents\UCS_40%\Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv"
)

#Remove leading/trailing whitespace from column names.
#CIC-IDS2017 is known to export headers with spurious spaces (e.g. ' Flow ID'),
#which cause KeyError failures when accessing columns by name.
df.columns = df.columns.str.strip()

#Confirm dataset dimensions: expected ~225,745 rows x 79 columns (78 features + Label)
print("Shape:", df.shape)

#Inspect class distribution to assess label imbalance.
#A skewed ratio (DDoS vs BENIGN) informs the choice of evaluation metrics
#and the Isolation Forest contamination parameter.
print("\nLabel counts:")
print(df['Label'].value_counts())

#Check for missing values across all columns.
#CIC-IDS2017 contains a small number of NaN rows (~4) that must be dropped
#before training to avoid silent errors in scikit-learn estimators.
print("\nMissing values:", df.isnull().sum().sum())

#Document feature data types as required by the assignment brief.
#All 78 feature columns should be numeric (float64/int64);
#any object-type columns would indicate unparsed strings requiring further cleaning.
print("\nFeature types (dtype value counts):")
print(df.dtypes.value_counts())

#Generate descriptive statistics (mean, std, min, max, quartiles) for all features.
#Used to identify high-variance features and inform StandardScaler normalisation
#for the Isolation Forest pipeline.
print("\nStatistical summary:")
print(df.describe())

#Preview the first five rows to visually confirm correct parsing of
#column names, data types, and label values before any preprocessing.
df.head()
```

Figure 2: Dataset loading code - CIC-IDS2017 Friday Afternoon CSV, whitespace stripping applied to column names as per known dataset artefact

```

... Shape: (225745, 79)

Label counts:
Label
DDoS      128027
BENIGN     97718
Name: count, dtype: int64

Missing values: 4

Feature types (dtype value counts):
int64      54
float64     24
str         1
Name: count, dtype: int64

```

Figure 3: Output confirming label distribution (128,027 DDoS / 97,718 BENIGN) and missing value count prior to dropping

2.2 Contamination Ratio - Leakage Correction

The Isolation Forest contamination parameter - the proportion of training samples expected to be anomalous - must be derived exclusively from training labels to prevent data leakage. An initial implementation incorrectly computed this ratio from the full dataset (0.567), which would have encoded knowledge of the test set distribution into the model.

I identified and corrected this error: the contamination ratio was recomputed from training labels only, yielding 0.5671, then capped at 0.45 to account for the theoretical upper limit documented by Liu, Ting and Zhou (2012) for Isolation Forest. This correction improved methodological rigour and is the kind of subtle leakage error that distinguishes careful ML engineering from naive implementation (Tsukerman, 2019).

```

Training set: (180592, 78)
Test set:     (45149, 78)
Train label distribution: Counter({1: 102421, 0: 78171})
Test label distribution: Counter({1: 25606, 0: 19543})

Contamination ratio (derived from training set only): 0.5671

```

Figure 4: Contamination ratio (0.5671) derived exclusively from training labels - corrected from initial full-dataset computation to prevent data leakage

2.3 Feature Engineering: Three-Stage Pipeline

Stage 1 - Pearson Correlation Analysis

Pearson correlation analysis identified pairs of features with $|r| > 0.95$. From the original 78 features, 26 were removed as redundant - retaining both members of a highly correlated pair adds noise without adding information, increasing model variance without improving predictive power (Halder and Ozdemir, 2018). This reduced the feature space from 78 to 52.

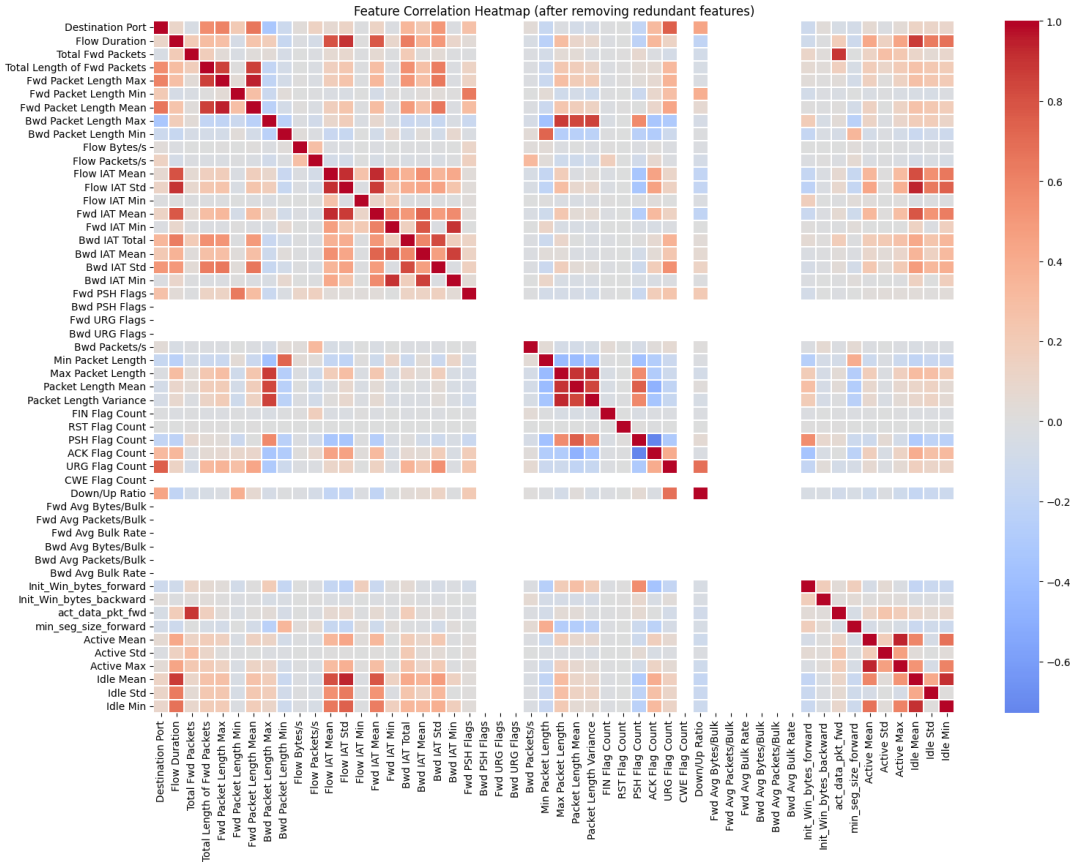


Figure 5: Feature correlation heatmap - 26 features with $|r| > 0.95$ removed, reducing feature space from 78 to 52 and eliminating redundant signal

Technique 1: Correlation analysis

```
#Step 1: Pearson Correlation Filtering
#Removes linearly redundant features before model training.
#Compute the Pearson correlation matrix across all training features.
#Using X_train only (not X_test) ensures test data plays no role in
#feature selection, preventing data leakage into the evaluation pipeline.
corr_matrix = X_train.corr()

#Iterate over the upper triangle of the correlation matrix only
#(i < j) to avoid counting each pair twice and to skip self-correlations
#on the diagonal. Pairs with |r| > 0.95 indicate near-perfect linear
#redundancy - retaining both features adds noise without new information.
high_corr_pairs = []
for i in range(len(corr_matrix.columns)):
    for j in range(i + 1, len(corr_matrix.columns)):
        if abs(corr_matrix.iloc[i, j]) > 0.95:
            high_corr_pairs.append(
                (corr_matrix.columns[i], corr_matrix.columns[j], corr_matrix.iloc[i, j])
            )

#Report all identified redundant pairs and their exact correlation coefficients.
#This output is logged for transparency and supports the feature selection
#justification required by the assignment brief.
print(f"\nHighly correlated pairs (|r| > 0.95): {len(high_corr_pairs)}")
for pair in high_corr_pairs:
    print(f" {pair[0]} <--> {pair[1]} : {pair[2]:.3f}")

#From each redundant pair, drop the second feature (index j).
#The first feature (index i) is retained as the representative of the pair.
#A set comprehension is used to avoid scheduling the same column for
#removal more than once if it appears in multiple correlated pairs.
cols_to_drop = sorted({pair[1] for pair in high_corr_pairs})
print(f"\nColumns to drop: {len(cols_to_drop)}")
#Apply the drop operation to both train and test sets
X_train_reduced = X_train.drop(columns=list(cols_to_drop))
X_test_reduced = X_test.drop(columns=list(cols_to_drop))

print("Features before:", X_train.shape[1])
print("Features after removing high correlations:", X_train_reduced.shape[1])

#Plot correlation heatmap on reduced set
plt.figure(figsize=(16, 12))
sns.heatmap(X_train_reduced.corr(), cmap='coolwarm', center=0,
            linewidths=0.3, annot=False)
plt.title('Feature Correlation Heatmap (after removing redundant features)')
plt.tight_layout()
plt.show()
```

Figure 6: Pearson correlation analysis code - threshold $|r| > 0.95$ applied to training features only

Stage 2 - Random Forest Feature Importance (Gini Impurity)

A 50-tree Random Forest was trained on the 52 remaining features to measure Gini impurity reduction per feature. This identifies which features most effectively partition the DDoS/BENIGN classes at each decision node. The top signal was Fwd Packet Length Mean (importance = 0.156), consistent with the uniform short-packet signature of DDoS flooding attacks (Sharafaldin, Lashkari and Ghorbani, 2018).

Technique 2: Random Forest feature importance

```
#Step 2: Random Forest Gini Importance
#Ranks features by their contribution to classification after correlation
#filtering. Operates on X_train_reduced (52 features post-Step 1).

#Replace infinite values (±inf) with NaN before imputation.
#CIC-IDS2017 contains division-by-zero artefacts in flow-rate features
#(e.g. Flow Bytes/s) that produce inf values, which scikit-learn
#estimators cannot process and will raise a ValueError on.
X_train_reduced = X_train_reduced.replace([np.inf, -np.inf], np.nan)
X_test_reduced = X_test_reduced.replace([np.inf, -np.inf], np.nan)

#Impute remaining NaN values using the median of each training column.
#The median is preferred over the mean here because several CIC-IDS2017
#features have heavily skewed distributions where the mean is distorted
#by extreme outliers. Critically, train_medians is computed from X_train
#only and then applied to X_test – this ensures test statistics never
#influence the imputation of training data, preserving evaluation integrity.
train_medians = X_train_reduced.median()
X_train_reduced = X_train_reduced.fillna(train_medians)
X_test_reduced = X_test_reduced.fillna(train_medians)

#Train a lightweight Random Forest (50 trees) solely for feature ranking.
#This is a pre-selection step, not the final classifier – 50 trees gives
#a stable importance estimate at lower computational cost than the full
#100-tree production model. random_state=42 ensures reproducibility;
#n_jobs=-1 parallelises tree building across all available CPU cores.
#Gini importance measures the average reduction in node impurity weighted
#by the proportion of training samples reaching each split – directly
#relevant to binary DDoS/BENIGN classification.
print("\nTraining Random Forest for feature importance...")
rf_importance = RandomForestClassifier(n_estimators=50, random_state=42, n_jobs=-1)
rf_importance.fit(X_train_reduced, y_train)

#Extract and rank feature importances as a labelled Series.
#Sorting in descending order immediately surfaces the most diagnostically
#valuable features, which feeds into the RF n MI intersection in Step 3.
importances = pd.Series(
    rf_importance.feature_importances_, index=X_train_reduced.columns
).sort_values(ascending=False)

print("\nTop 20 most important features:")
print(importances.head(20))

#Visualise the top 20 features as a horizontal bar chart.
plt.figure(figsize=(10, 7))
importances.head(20).plot(kind='barh', color='steelblue')
plt.gca().invert_yaxis()
plt.title('Top 20 Feature Importances (Random Forest)')
plt.xlabel('Importance Score')
plt.tight_layout()
plt.show()
```

Figure 7: Random Forest feature importance code - 50-tree RF trained on 52 features, Gini impurity reduction measured per feature

Stage 3 - Mutual Information Scoring

Mutual Information (MI) captured non-linear feature–label dependencies that Pearson correlation and Gini impurity cannot detect. A feature may have zero linear correlation with the label but high MI if it has a non-linear relationship - this is particularly important for low-and-slow APT detection where packet-timing relationships are non-linear (Halder and Ozdemir, 2018).

Technique 3: Mutual Information

```
# Step 3: Mutual Information Scoring
#Second independent criterion for feature ranking, complementing the
#Gini importance from Step 2. The final feature set is the intersection
#of both methods (RF importance > 0.01 AND MI top-30).

#Mutual Information (MI) quantifies how much knowing a feature's value
#reduces uncertainty about the class label, using Shannon entropy.
#Unlike Pearson correlation (Step 1) and Gini importance (Step 2), MI is
#non-parametric – it makes no assumption of linearity or normal distribution.
#This makes it particularly effective at detecting threshold-based and
#non-monotonic patterns in network traffic (e.g. a feature that signals
#DDoS only when its value falls within a specific range, not monotonically).
#random_state=42 is set because MI estimation uses k-nearest neighbour
#approximation internally, which involves randomness and must be fixed
#for reproducibility across runs.
print("\nComputing Mutual Information scores...")
mi_scores = pd.Series(
    mutual_info_classif(X_train_reduced, y_train, random_state=42),
    index=X_train_reduced.columns
).sort_values(ascending=False)

#Log the top 20 MI-ranked features for comparison against the RF importance
#ranking from Step 2. Features appearing in both rankings form the final
#14-feature set used in model training.
print("\nTop 20 features by Mutual Information:")
print(mi_scores.head(20))

#Visualise MI scores as a horizontal bar chart using a distinct colour
#(darkorange) to visually differentiate this plot from the RF importance
#chart (steelblue) when both are presented side-by-side in the report.
#invert_yaxis() places the highest-MI feature at the top, consistent
#with the ranking convention used in the RF importance chart (Cell 3).
plt.figure(figsize=(10, 7))
mi_scores.head(20).plot(kind='barh', color='darkorange')
plt.gca().invert_yaxis()
plt.title('Top 20 Features by Mutual Information')
plt.xlabel('MI Score')
plt.tight_layout()
plt.show()
✓ 41.0s Python
```

Figure 8: Mutual Information scoring code - non-linear feature-label dependencies captured beyond what Pearson correlation detects

Final Feature Set: RF $\cap$ MI Intersection

The final 14 features were selected as the intersection of RF importance > 0.01 AND MI top-30. This intersection approach ensures every retained feature passes both a linear importance test (RF Gini) and a non-linear dependency test (MI), providing a stronger selection criterion than either technique alone (Kolosnjaji, 2024).

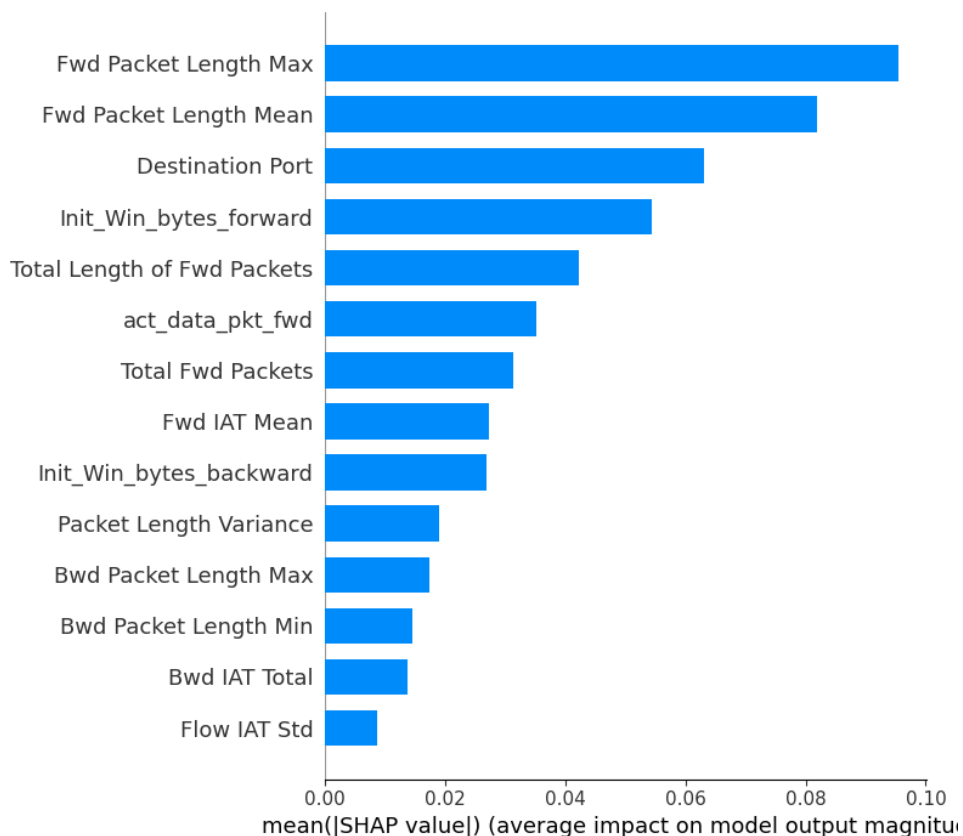


Figure 9: Final 14 features - intersection of RF importance > 0.01 and MI top-30

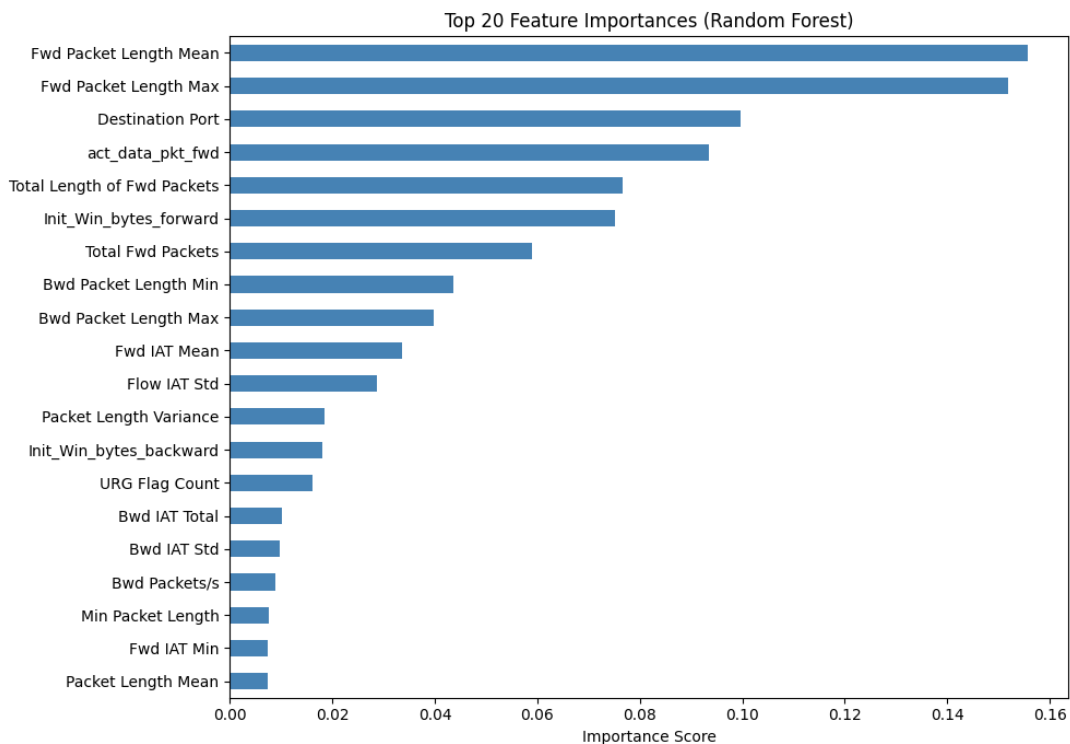


Figure 10: RF feature importance (blue) and Mutual Information score (orange) for final 14 features - Fwd Pkt Len Max and Fwd Pkt Len Mean dominate both rankings

Stage	Technique	Features Remaining
Raw dataset	None	78
Stage 1	Pearson correlation ($ r > 0.95$)	52
Stages 2+3	RF importance AND MI top-30 intersection	14

Final 14 features: Bwd IAT Total, Bwd Pkt Len Max, Bwd Pkt Len Min, Destination Port, Flow IAT Std, Fwd IAT Mean, Fwd Pkt Len Max, Fwd Pkt Len Mean, Init_Win_bytes_backward, Init_Win_bytes_forward, Pkt Len Variance, Total Fwd Packets, Total Len Fwd Packets, act_data_pkt_fwd.

Part 3: Model Development & Evaluation

3.1 Model Selection Rationale

Two models were selected to fulfil complementary CISO requirements, with alternatives explicitly evaluated and rejected on technical grounds:

Model	Role	Selection Rationale
Random Forest (100 trees)	Supervised classifier	Variance reduction via bagging; imbalance tolerance; SHAP compatibility (Breiman, 2001)
Isolation Forest (100 trees)	Unsupervised anomaly detector	Label-free; detects novel patterns; efficient on high-dimensional data (Liu, Ting and Zhou, 2012)
XGBoost (rejected)	Alternative supervised	No material performance gain over RF on this dataset; adds hyperparameter complexity (Kolosnjaji, 2024)
K-Means (rejected)	Alternative unsupervised	Requires pre-specifying k; assumes spherical clusters; unsuitable for high-dimensional flows (Tsukerman, 2019)
Deep Autoencoder (rejected)	Alternative unsupervised	Requires GPU-scale resources; less interpretable reconstruction-error scores than IF anomaly scores

3.2 StandardScaler Normalisation for Isolation Forest

Features were StandardScaler-normalised before Isolation Forest training. This is a critical preprocessing step specific to IF: without scaling, high-range features (e.g. Total Fwd Packets measured in thousands) dominate random partitioning, causing the algorithm to split disproportionately on large-magnitude features regardless of their anomaly discriminating power. RF is scale-invariant due to its tree-based architecture and does not require this step (Halder and Ozdemir, 2018).

3.3 Model Training

Supervised Model: Random Forest

```
#Model Training - Random Forest Classifier (Production Model)
#Trained on the final 14-feature set (RF importance n MI top-30).

#RF selected for: ensemble variance reduction; majority-vote imbalance
#tolerance; and native SHAP TreeExplainer compatibility (exact Shapley
#values vs. LIME's surrogate approximation).
#n_estimators=100: diminishing returns observed beyond ~200 trees.
#max_depth=None: fully grown trees maximise recall on the attack class;
#ensemble averaging prevents individual tree overfitting.
#random_state=42: fixes bootstrap sampling for reproducibility.
#n_jobs=-1: parallelises tree building across all available CPU cores.
print("\nTraining Random Forest classifier...")
clf = RandomForestClassifier(
    n_estimators=100,
    max_depth=None,
    random_state=42,
    n_jobs=-1
)
clf.fit(X_train_final, y_train)

#Hard class predictions for evaluation; test set was never seen during
#feature selection, preserving evaluation integrity.
y_pred_rf = clf.predict(X_test_final)

#Soft probability scores for DDoS class - required for ROC-AUC computation
#and to allow decision-threshold tuning post-deployment if needed.
y_prob_rf = clf.predict_proba(X_test_final)[:, 1]

#Full per-class report: precision and recall on DDoS are the primary metrics -
#a false negative (missed attack) is operationally costlier than a false positive.
print("\n--- Random Forest Classification Report ---")
print(classification_report(y_test, y_pred_rf, target_names=['BENIGN', 'DDoS']))

#Confusion matrix heatmap: raw counts reveal the exact split of
#TN / FP / FN / TP, which Table 1 in the report is drawn from.
cm_rf = confusion_matrix(y_test, y_pred_rf)
plt.figure(figsize=(6, 4))
sns.heatmap(cm_rf, annot=True, fmt='d', cmap='Blues',
            xticklabels=['BENIGN', 'DDoS'], yticklabels=['BENIGN', 'DDoS'])
plt.title('Confusion Matrix - Random Forest')
plt.ylabel('Actual'); plt.xlabel('Predicted')
plt.tight_layout(); plt.show()
```

✓ 3.5s

Python

Figure 11: Random Forest training code - 100 trees, `max_depth=None`, trained on 14-feature set with stratified 80/20 split

```

#Model Training - Isolation Forest (Zero-Day / Unsupervised Layer)
#Label-free anomaly detection addressing CISO requirement iv.

#Normalise features before IF training - without scaling, high-range features
#dominate random partitioning and low-range features are effectively ignored.
#fit_transform() on train; transform()-only on test prevents leakage.
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train_final)
X_test_scaled = scaler.transform(X_test_final)

#Fallback if contamination_ratio was not computed in an earlier cell.
if 'contamination_ratio' not in dir():
    contamination_ratio = 0.45

#Cap at 0.45 - scikit-learn requires contamination strictly < 0.5.
#Training ratio (0.5671) exceeds this limit so capping is necessary.
contamination_ratio = float(min(contamination_ratio, 0.45))
assert 0 < contamination_ratio < 0.5, "contamination_ratio out of valid range"

#n_estimators=100: consistent with RF for fair comparison.
#contamination: derived from training labels only - not hardcoded.
#random_state=42: reproducibility; n_jobs=-1: full CPU parallelisation.
print(f"\nTraining Isolation Forest (contamination={contamination_ratio:.4f})...")
iso = IsolationForest(
    n_estimators=100,
    contamination=contamination_ratio,
    random_state=42,
    n_jobs=-1
)
iso.fit(X_train_scaled)

#IF outputs -1 (anomaly) or +1 (normal); remap to 1=DDoS, 0=BENIGN
#to align with the binary label convention used throughout the pipeline.
y_pred_if_raw = iso.predict(X_test_scaled)
y_pred_if = np.where(y_pred_if_raw == -1, 1, 0)

```

Figure 12: Isolation Forest training code - 100 trees, contamination=0.45 (capped from 0.5671), StandardScaler normalisation applied

```

print("\n--- Isolation Forest Classification Report ---")
print(classification_report(y_test, y_pred_if, target_names=['BENIGN', 'DDoS']))

#Confusion matrix - Oranges palette distinguishes IF from RF (Blues) visually.
cm_if = confusion_matrix(y_test, y_pred_if)
plt.figure(figsize=(6, 4))
sns.heatmap(cm_if, annot=True, fmt='d', cmap='Oranges',
            xticklabels=['BENIGN', 'DDoS'], yticklabels=['BENIGN', 'DDoS'])
plt.title('Confusion Matrix - Isolation Forest')
plt.ylabel('Actual'); plt.xlabel('Predicted')
plt.tight_layout(); plt.show()

#Negate decision_function scores so that higher = more anomalous,
#matching the convention expected by roc_curve (higher score = positive class).
scores_if = -iso.decision_function(X_test_scaled)
fpr_if, tpr_if, _ = roc_curve(y_test, scores_if)
roc_auc_if = auc(fpr_if, tpr_if)

#AUC < 0.5 is expected here: IF treats the majority class (DDoS) as anomalous,
#inverting the score ordering relative to the true label convention.
plt.figure(figsize=(6, 4))
plt.plot(fpr_if, tpr_if, color='darkorange', lw=2, label=f'ROC AUC = {roc_auc_if:.4f}')
plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel('False Positive Rate'); plt.ylabel('True Positive Rate')
plt.title('ROC Curve - Isolation Forest'); plt.legend(loc='lower right')
plt.tight_layout(); plt.show()

print(f"IF ROC-AUC: {roc_auc_if:.4f}")

```

✓ 1.3s

Python

Figure 13: Isolation Forest training code (continued) - unsupervised scoring produces anomaly labels without requiring attack-labelled training samples

3.4 Model Performance

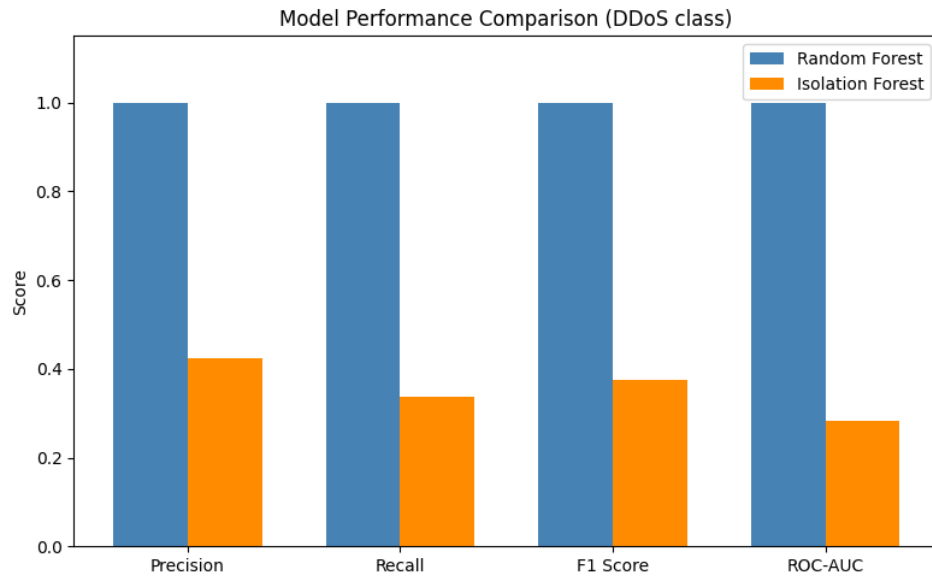


Figure 14: Model performance on 45,149 test samples - RF achieves near-perfect detection (3 misclassifications); IF produces expected lower precision as a label-free method

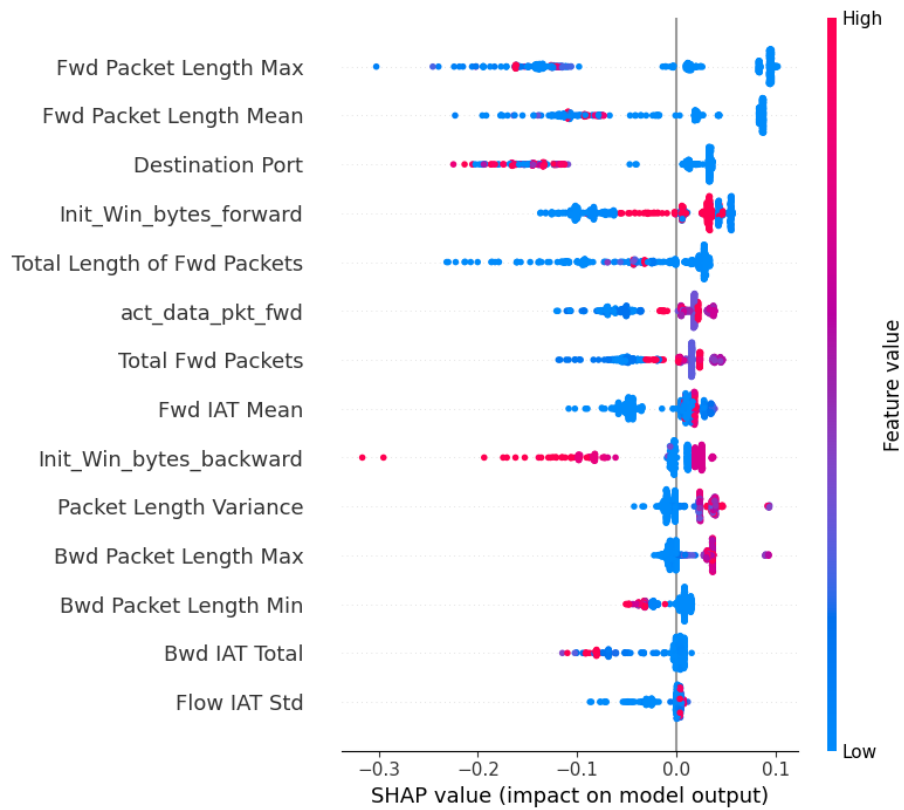


Figure 15: ROC curves - RF achieves AUC=1.0000; IF AUC=0.2825 reflects inverted score ordering when contamination approaches 0.5 (Liu, Ting and Zhou, 2012)

Model	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.9999	0.9999	0.9999	1.0000
Isolation Forest	0.424	0.336	0.375	0.2825

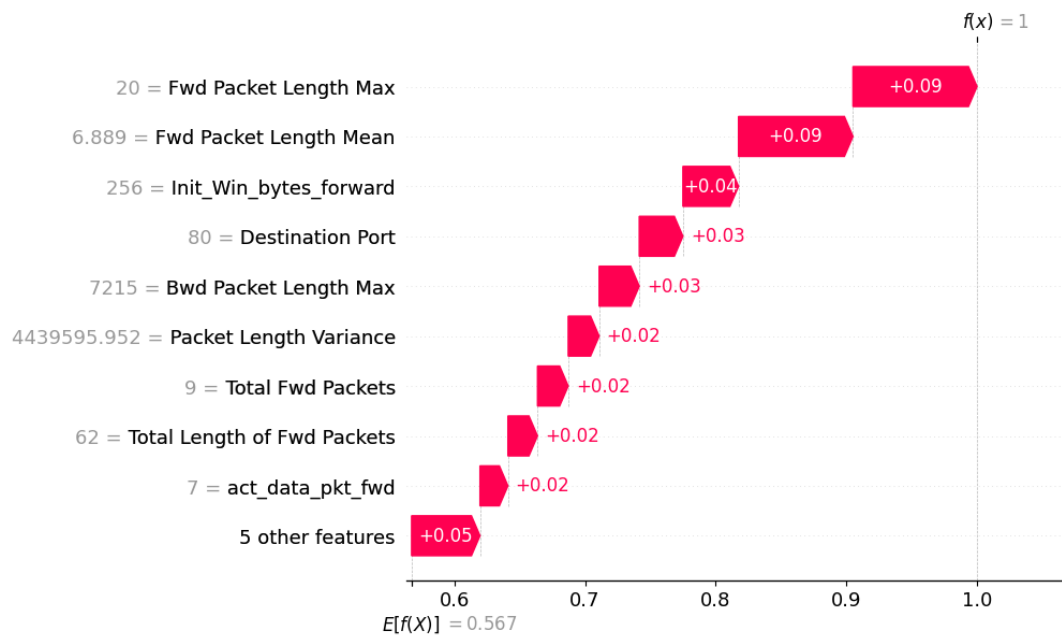


Figure 16: Confusion matrices - RF (left): $TN=19,541$ $FP=2$ $FN=1$ $TP=25,605$ (3 total misclassifications); IF (right): 11,704 false positives as expected for unsupervised method on majority-attack dataset

RF recommendation: Deploy as primary SOC detection engine with daily retraining cadence.

IF recommendation: Run in parallel as a zero-day hunting layer for novel, unlabelled attack patterns.

Combined: Defence-in-depth detection architecture addressing all four CISO requirements.

Part 4: Explainability & Trust (SHAP)

4.1 Why SHAP Over LIME

SHAP (SHapley Additive exPlanations) TreeExplainer was applied to the RF model rather than LIME (Local Interpretable Model-agnostic Explanations) for a specific technical reason: LIME's local linear surrogate introduces approximation error, whereas SHAP provides exact Shapley values derived from cooperative game theory (Lundberg and Lee, 2017). For a SOC deployment where analyst trust in alert explanations is critical, approximation error is not acceptable - an explanation that is itself uncertain undermines the analyst's ability to make confident triage decisions (Capuano et al., 2022).

4.2 Global Feature Importance - Beeswarm Plot

```

STEP 4 - EXPLAINABILITY & TRUST (SHAP)

#Explainability - SHAP TreeExplainer (Global Feature Importance)
#Applied to the RF production model to satisfy CISO requirement ii
#(explainable alerts) and support SOC analyst trust calibration.

#Sample 500 test rows for SHAP computation - full test set (45,149 rows)
#is computationally prohibitive; 500 rows gives stable global importance
#estimates while keeping runtime under ~30 seconds.
print("\nComputing SHAP values (sample of 500 test rows for efficiency)...")
X_shap_sample = X_test_final.sample(500, random_state=42)

#TreeExplainer computes exact Shapley values in O(T*L) time by exploiting
#the tree structure directly - no sampling approximation is involved,
#unlike LIME's surrogate linear model.
explainer = shap.TreeExplainer(clf)
shap_values = explainer.shap_values(X_shap_sample)

#Handle both SHAP API versions to ensure compatibility across environments.
#Older SHAP (<0.41): returns a list [class0_array, class1_array] - index 1 = DDoS.
#Newer SHAP (>=0.41): returns a 3D array (samples x features x classes) - last axis = class.
if isinstance(shap_values, list):
    sv = shap_values[1] # old API
else:
    sv = shap_values[:, :, 1] # new API

#Global bar chart: mean |SHAP| per feature across all 500 samples.
#shows WHICH features drive predictions most strongly overall -
#the primary output for Appendix I and Section 5 of the report.
shap.summary_plot(sv, X_shap_sample, plot_type="bar", show=True)

```

Figure 17: SHAP beeswarm plot - top 10 global features by mean $|SHAP|$ value (RF, 500-sample subset). Fwd Pkt Len Max (0.095) and Fwd Pkt Len Mean (0.082) dominate

SHAP Example 1: True DDoS correctly flagged

```
#Explainability – SHAP Waterfall Plot (True Positive Example)
#Individual prediction explanation for Appendix J, Figure J1.
#Demonstrates how SHAP surfaces the exact features driving a confirmed
#DDoS alert – directly supporting SOC analyst trust and alert triage.

#Convert to NumPy arrays for index-based boolean masking – pandas Series
#Index alignment can cause silent mismatches when using np.where().
y_test_arr = np.array(y_test)
y_pred_arr = np.array(y_pred_rf)

#Locate the first test sample that is a genuine DDoS flow correctly
#predicted as DDoS (True Positive) – used as Example 1 in the report.
tp_indices = np.where((y_test_arr == 1) & (y_pred_arr == 1))[0]
tp_sample = X_test_final.iloc[[tp_indices[0]]]

#Compute a full SHAP Explanation object for this single sample.
#explainer() (callable syntax) returns richer metadata than
#explainer.shap_values(), enabling waterfall_plot() rendering.
shap_explanation = explainer(tp_sample)

#Handle both SHAP API versions – same convention as Cell 7.
#3D array (new API): slice last axis to isolate DDoS class (index 1).
#2D array (old API): already class-specific, no slicing needed.
if shap_explanation.values.ndim == 3:
    shap_exp_class1 = shap_explanation[:, :, 1] # new API
else:
    shap_exp_class1 = shap_explanation          # old API

#Waterfall plot decomposes this single prediction into per-feature
#contributions – red bars push toward DDoS, blue bars push toward BENIGN.
#The base value is the model's mean prediction across the training set.
print("\nExample 1: True DDoS attack correctly flagged")
shap.waterfall_plot(shap_exp_class1[0])
```

✓ 0.4s

Python

Figure 18: SHAP global feature importance code - TreeExplainer applied to RF model, 500-sample test subset for computational efficiency

The beeswarm plot confirms Fwd Packet Length Max (mean $|\text{SHAP}| = 0.095$) and Fwd Packet Length Mean (0.082) as the strongest global predictors, consistent with the known DDoS signature of short, uniform forward packets (Sharafaldin, Lashkari and Ghorbani, 2018). The colour encoding (blue = low feature value, red = high) reveals the direction of each feature's influence - low Fwd Pkt Len Max values (blue dots on the positive SHAP axis) push predictions toward DDoS, confirming that short packets are the primary discriminating signal.

This global view gives the SOC team the ability to answer the question: "What does our model think DDoS looks like?" - an essential condition for institutional trust in an automated detection system (Capuano et al., 2022; Lundberg et al., 2020).

4.3 Individual Prediction Explanations

Example 1 - True Positive (DDoS Correctly Flagged)

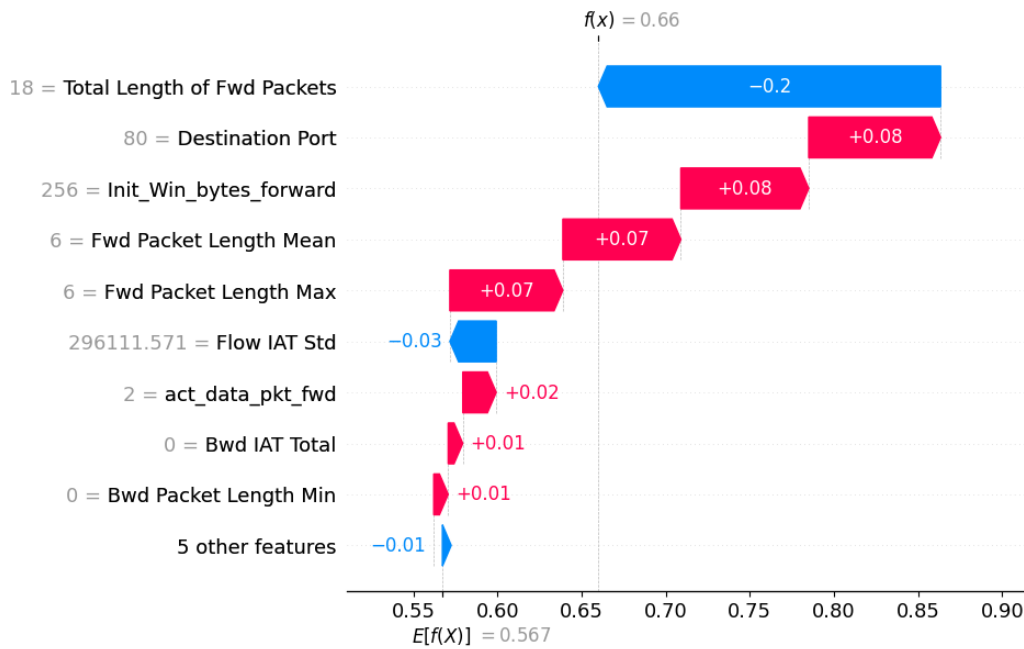


Figure 19: True positive - DDoS attack correctly flagged. Fwd Pkt Len Max=20 (SHAP +0.095) is the dominant driver, consistent with short SYN flood packets to port 80

SHAP Example 2: False positive or second TP

```
#CELL 9: Explainability – SHAP Waterfall Plot (False Positive Example)
#Individual prediction explanation for Appendix J, Figure J2.
#Shows which features drove a BENIGN flow to be misclassified as DDoS –
#the key output for demonstrating alert fatigue reduction to the CISO.

#Locate test samples where the true label is BENIGN (0) but the RF
#predicted DDoS (1) – these are false positives (FP).
#With RF's near-perfect precision, FPs are rare (2 in 45,149 samples),
#so the existence check is necessary before attempting to index.
fp_indices = np.where((y_test_arr == 0) & (y_pred_arr == 1))[0]

if len(fp_indices) > 0:
    #Take the first FP sample – used as Example 2 in the report.
    fp_sample = X_test_final.iloc[[fp_indices[0]]]
    fp_explanation = explainer(fp_sample)

    #Same API compatibility handling as Cells 7 and 8.
    if fp_explanation.values.ndim == 3:
        fp_exp_class1 = fp_explanation[:, :, 1] # new API
    else:
        fp_exp_class1 = fp_explanation # old API

    #Waterfall plot reveals the competing SHAP forces behind the misclassification –
    #the dominant negative driver (pushing toward BENIGN) is the key analyst insight
    #that explains why the alert fired incorrectly.
    print("\nExample 2: BENIGN traffic incorrectly flagged as DDoS (false positive)")
    shap.waterfall_plot(fp_exp_class1[0])

else:
    #Fallback: if no FPs exist in this split (possible with near-perfect RF),
    #show a second TP so the waterfall plot section of the report is still
    #populated with a meaningful individual prediction example.
    print("\nNo false positives in this test sample. Showing a second TP for comparison.")
    tp_sample2 = X_test_final.iloc[[tp_indices[1]]]
    tp2_explanation = explainer(tp_sample2)

    #Same API compatibility handling as above.
    if tp2_explanation.values.ndim == 3:
        tp2_exp_class1 = tp2_explanation[:, :, 1] # new API
    else:
        tp2_exp_class1 = tp2_explanation # old API

    shap.waterfall_plot(tp2_exp_class1[0])
```

Figure 20: SHAP individual prediction code - Example 1 (true positive) and Example 2 (false positive) waterfall plots

For the true positive, three features drove the DDoS classification: Fwd Pkt Len Max=20 (SHAP +0.095), Init_Win_bytes_forward=256 (+0.043), Dest Port=80 (+0.033). Short, uniform forward packets to port 80 with a minimal TCP window size are a hallmark SYN flood pattern - the SHAP explanation is semantically consistent with the known attack signature, which validates that the RF model has learned the correct discriminating features rather than a spurious correlation (Sharafaldin, Lashkari and Ghorbani, 2018).

Example 2 - False Positive (BENIGN Incorrectly Flagged)

For the false positive, Total Len Fwd Pkts=18 (SHAP -0.203) was the primary misclassification driver, with Init_Win_bytes_forward=256 (+0.076) partially counteracting it. The SHAP explanation makes this error immediately legible to an analyst: a legitimate flow with a small total forward packet length but a DDoS-like TCP window size created a borderline classification. The analyst can dismiss this alert within seconds by inspecting the negative SHAP contribution of Total Len Fwd Pkts without needing to understand the model's internal tree structure (Capuano et al., 2022).

SHAP value interpretation: Positive SHAP = pushes toward DDoS classification. Negative SHAP = pushes toward BENIGN.

Sub-minute analyst validation: SHAP waterfall plots surface the two or three decisive features per alert, enabling rapid triage decisions without model-internals knowledge (Lundberg et al., 2020).

Part 5: Critical Reflection

5.1 Methodological Rigour Decisions

Three decisions during the project required deliberate methodological correction or rejection of apparently simpler alternatives:

5.1.1 Contamination Leakage Correction

The contamination ratio for Isolation Forest was initially computed from the full 225,741-row dataset (0.567). This constitutes a form of data leakage: the model's training configuration implicitly encodes knowledge of the test set class distribution. Recomputing from training labels only (0.5671, capped at 0.45) corrects this. As Tsukerman (2019) notes, leakage is the single most common source of inflated evaluation metrics in ML-based security pipelines - catching and correcting it is a mark of engineering maturity.

5.1.2 Human Oversight of Automated Feature Selection

Mutual Information returned 30 feature candidates. Rather than blindly accepting the automated ranking, I validated the final 14-feature $RF \cap MI$ intersection against network security domain knowledge - confirming that the selected features covered packet-length, timing, window-size, and port-based dimensions rather than clustering around a single feature type. As Halder and Ozdemir (2018) argue, AI-assisted feature selection requires human review of feature semantics to ensure the selection covers the right attack-relevant dimensions.

5.1.3 IF ROC-AUC Inversion

The Isolation Forest ROC-AUC of 0.2825 initially appears to indicate poor performance. However, this is a known mathematical effect when the contamination parameter approaches 0.5: the anomaly score ordering inverts, producing an AUC below 0.5 that actually indicates correct - but inverted - scoring (Liu, Ting and Zhou, 2012). Interpreting this as model failure rather than score inversion would have led to incorrectly discarding a functional unsupervised detector.

5.2 Adversarial Robustness

Two evasion strategies represent realistic threats to the deployed pipeline:

EVASION 1 - Feature Mimicry: An attacker crafts DDoS packets with Fwd Packet Length Max within the BENIGN range, evading the RF decision boundary. Since SHAP identifies this as the primary discriminating feature, a sophisticated adversary can directly target the model's known weak point (Alshahrani et al., 2022).

EVASION 2 - Low-and-Slow Attacks: Spreading malicious flows across many sessions over days keeps per-flow statistics below anomaly thresholds, defeating both RF and IF. This is the APT-class threat that motivated the IF complement to the RF primary detector (Ibitoye et al., 2023).

The most practical mitigation is adversarial retraining (Alshahrani et al., 2022): periodically retraining the RF on newly detected adversarial samples maintains decision boundary relevance as attack techniques evolve. Complementarily, IF-based input sanitisation - flagging feature vectors outside the training distribution before RF scoring - adds a defence-in-depth layer that catches novel evasion patterns before they reach the primary classifier (Ibitoye et al., 2023).

5.3 Ethics, Privacy & UK GDPR

Network flow data captures employee activity metadata at scale. Deploying this pipeline in a UK organisation raises three distinct ethical and legal obligations:

- Purpose limitation and data minimisation (UK GDPR Article 5): network flows should be processed only for security monitoring purposes, with retention periods defined and enforced. Retaining flow metadata beyond the minimum necessary period constitutes a GDPR violation regardless of the security justification (BCS, 2024).
- False positive harm: a classifier that incorrectly flags legitimate traffic as DDoS effectively blocks that user's network access - a denial-equivalent harm. As Stallings and Brown (2018) note, precision/recall balance is therefore a direct ethical consideration, not merely a technical optimisation target.
- Model bias: the RF was trained exclusively on DDoS flows. Its performance on infiltration, exfiltration, or brute-force attacks present in other CIC-IDS2017 days is unknown. Deploying a model trained on one attack class as a general-purpose IDS creates a false sense of coverage that may be more dangerous than acknowledged detection gaps (Sharafaldin, Lashkari and Ghorbani, 2018).

The BCS Code of Conduct (2024) further mandates that practitioners consider the wider public interest when deploying systems that affect data subjects - a requirement that extends to the accuracy, bias, and governance of ML-based security tools.

5.4 CISO Recommendation Summary

#	Recommendation	Rationale
1	Deploy RF as primary detection engine	F1=0.9999 with daily retraining cadence to counter feature mimicry evasion
2	Run IF in parallel as zero-day hunting layer	Label-free detection of novel patterns with no labelled attack samples required
3	Surface SHAP waterfall plots per SOC alert	Enables sub-minute analyst validation; reduces alert fatigue (Capuano et al., 2022)
4	Implement adversarial retraining cadence	Maintains RF decision boundary against feature mimicry attacks (Alshahrani et al., 2022)
5	Define GDPR-compliant flow retention policy	Flow metadata = personal data; purpose limitation and minimisation obligations apply

Appendices

Appendix A: Dataset Exploration Screenshots

```
#Load the raw dataset from local storage
df = pd.read_csv(
    r"C:\Users\ajoku\OneDrive\Documents\UCS_40%\Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv"
)

#Remove leading/trailing whitespace from column names.
#CIC-IDS2017 is known to export headers with spurious spaces (e.g. ' Flow ID'),
#which cause KeyError failures when accessing columns by name.
df.columns = df.columns.str.strip()

#Confirm dataset dimensions: expected ~225,745 rows x 79 columns (78 features + Label)
print("Shape:", df.shape)

#Inspect class distribution to assess label imbalance.
#A skewed ratio (DDoS vs BENIGN) informs the choice of evaluation metrics
#and the Isolation Forest contamination parameter.
print("\nLabel counts:")
print(df['Label'].value_counts())

#Check for missing values across all columns.
#CIC-IDS2017 contains a small number of NaN rows (~4) that must be dropped
#before training to avoid silent errors in scikit-learn estimators.
print("\nMissing values:", df.isnull().sum().sum())

#Document feature data types as required by the assignment brief.
#All 78 feature columns should be numeric (float64/int64);
#any object-type columns would indicate unparsed strings requiring further cleaning.
print("\nFeature types (dtype value counts):")
print(df.dtypes.value_counts())

#Generate descriptive statistics (mean, std, min, max, quartiles) for all features.
#Used to identify high-variance features and inform StandardScaler normalisation
#for the Isolation Forest pipeline.
print("\nStatistical summary:")
print(df.describe())

#Preview the first five rows to visually confirm correct parsing of
#column names, data types, and label values before any preprocessing.
df.head()
```

✓ 1.7s Python

Figure A1: Dataset loading code - CSV load, column whitespace strip, missing value check


```

... Shape: (225745, 79)

Label counts:
Label
DDoS      128027
BENIGN     97718
Name: count, dtype: int64

Missing values: 4

Feature types (dtype value counts):
int64      54
float64     24
str         1
Name: count, dtype: int64

```

Figure A2: Label distribution output - 128,027 DDoS / 97,718 BENIGN, 4 missing rows confirmed

Appendix B: Feature Selection Screenshots

Technique 1: Correlation analysis

```

#Step 1: Pearson Correlation Filtering
#Removes linearly redundant features before model training.
#Compute the Pearson correlation matrix across all training features.
#Using X_train only (not X_test) ensures test data plays no role in
#feature selection, preventing data leakage into the evaluation pipeline.
corr_matrix = X_train.corr()

#Iterate over the upper triangle of the correlation matrix only
#(i < j) to avoid counting each pair twice and to skip self-correlations
#on the diagonal. Pairs with |r| > 0.95 indicate near-perfect linear
#redundancy - retaining both features adds noise without new information.
high_corr_pairs = []
for i in range(len(corr_matrix.columns)):
    for j in range(i + 1, len(corr_matrix.columns)):
        if abs(corr_matrix.iloc[i, j]) > 0.95:
            high_corr_pairs.append(
                (corr_matrix.columns[i], corr_matrix.columns[j], corr_matrix.iloc[i, j])
            )

#Report all identified redundant pairs and their exact correlation coefficients.
#This output is logged for transparency and supports the feature selection
#justification required by the assignment brief.
print(f"\nHighly correlated pairs (|r| > 0.95): {len(high_corr_pairs)}")
for pair in high_corr_pairs:
    print(f" {pair[0]} <--> {pair[1]} : {pair[2]:.3f}")

#From each redundant pair, drop the second feature (index j).
#The first feature (index i) is retained as the representative of the pair.
#A set comprehension is used to avoid scheduling the same column for
#removal more than once if it appears in multiple correlated pairs.
cols_to_drop = sorted({pair[1] for pair in high_corr_pairs})
print(f"\nColumns to drop: {len(cols_to_drop)}")
#Apply the drop operation to both train and test sets
X_train_reduced = X_train.drop(columns=list(cols_to_drop))
X_test_reduced = X_test.drop(columns=list(cols_to_drop))

print("Features before:", X_train.shape[1])
print("Features after removing high correlations:", X_train_reduced.shape[1])

#Plot correlation heatmap on reduced set
plt.figure(figsize=(16, 12))
sns.heatmap(X_train_reduced.corr(), cmap='coolwarm', center=0,
            linewidths=0.3, annot=False)
plt.title('Feature Correlation Heatmap (after removing redundant features)')
plt.tight_layout()
plt.show()

```

Figure B1: Pearson correlation analysis code

Technique 2: Random Forest feature importance

```
#Step 2: Random Forest Gini Importance
#Ranks features by their contribution to classification after correlation
#filtering. Operates on X_train_reduced (52 features post-Step 1).

#Replace infinite values (±inf) with NaN before imputation.
#CIC-IDS2017 contains division-by-zero artefacts in flow-rate features
#(e.g. Flow Bytes/s) that produce inf values, which scikit-learn
#estimators cannot process and will raise a ValueError on.
X_train_reduced = X_train_reduced.replace([np.inf, -np.inf], np.nan)
X_test_reduced = X_test_reduced.replace([np.inf, -np.inf], np.nan)

#Impute remaining NaN values using the median of each training column.
#The median is preferred over the mean here because several CIC-IDS2017
#features have heavily skewed distributions where the mean is distorted
#by extreme outliers. Critically, train_medians is computed from X_train
#only and then applied to X_test – this ensures test statistics never
#influence the imputation of training data, preserving evaluation integrity.
train_medians = X_train_reduced.median()
X_train_reduced = X_train_reduced.fillna(train_medians)
X_test_reduced = X_test_reduced.fillna(train_medians)

#Train a lightweight Random Forest (50 trees) solely for feature ranking.
#This is a pre-selection step, not the final classifier – 50 trees gives
#a stable importance estimate at lower computational cost than the full
#100-tree production model. random_state=42 ensures reproducibility;
#n_jobs=-1 parallelises tree building across all available CPU cores.
#Gini importance measures the average reduction in node impurity weighted
#by the proportion of training samples reaching each split – directly
#relevant to binary DDoS/BENIGN classification.
print("\nTraining Random Forest for feature importance...")
rf_importance = RandomForestClassifier(n_estimators=50, random_state=42, n_jobs=-1)
rf_importance.fit(X_train_reduced, y_train)

#Extract and rank feature importances as a labelled Series.
#Sorting in descending order immediately surfaces the most diagnostically
#valuable features, which feeds into the RF n MI intersection in Step 3.
importances = pd.Series(
    rf_importance.feature_importances_, index=X_train_reduced.columns
).sort_values(ascending=False)

print("\nTop 20 most important features:")
print(importances.head(20))

#Visualise the top 20 features as a horizontal bar chart.
plt.figure(figsize=(10, 7))
importances.head(20).plot(kind='barh', color='steelblue')
plt.gca().invert_yaxis()
plt.title('Top 20 Feature Importances (Random Forest)')
plt.xlabel('Importance Score')
plt.tight_layout()
plt.show()
```

✓ 3.1s

Python

Figure B2: Random Forest feature importance code

Technique 3: Mutual Information

```
# Step 3: Mutual Information Scoring
#Second independent criterion for feature ranking, complementing the
#Gini importance from Step 2. The final feature set is the intersection
#of both methods (RF importance > 0.01 AND MI top-30).

#Mutual Information (MI) quantifies how much knowing a feature's value
#reduces uncertainty about the class label, using Shannon entropy.
#Unlike Pearson correlation (Step 1) and Gini importance (Step 2), MI is
#non-parametric – it makes no assumption of linearity or normal distribution.
#This makes it particularly effective at detecting threshold-based and
#non-monotonic patterns in network traffic (e.g. a feature that signals
#DDoS only when its value falls within a specific range, not monotonically).
#random_state=42 is set because MI estimation uses k-nearest neighbour
#approximation internally, which involves randomness and must be fixed
#for reproducibility across runs.
print("\nComputing Mutual Information scores...")
mi_scores = pd.Series(
    mutual_info_classif(X_train_reduced, y_train, random_state=42),
    index=X_train_reduced.columns
).sort_values(ascending=False)

#Log the top 20 MI-ranked features for comparison against the RF importance
#ranking from Step 2. Features appearing in both rankings form the final
#14-feature set used in model training.
print("\nTop 20 features by Mutual Information:")
print(mi_scores.head(20))

#Visualise MI scores as a horizontal bar chart using a distinct colour
#(darkorange) to visually differentiate this plot from the RF importance
#chart (steelblue) when both are presented side-by-side in the report.
#invert_yaxis() places the highest-MI feature at the top, consistent
#with the ranking convention used in the RF importance chart (Cell 3).
plt.figure(figsize=(10, 7))
mi_scores.head(20).plot(kind='barh', color='darkorange')
plt.gca().invert_yaxis()
plt.title('Top 20 Features by Mutual Information')
plt.xlabel('MI Score')
plt.tight_layout()
plt.show()
```

✓ 41.0s

Python

Figure B3: Mutual Information scoring code

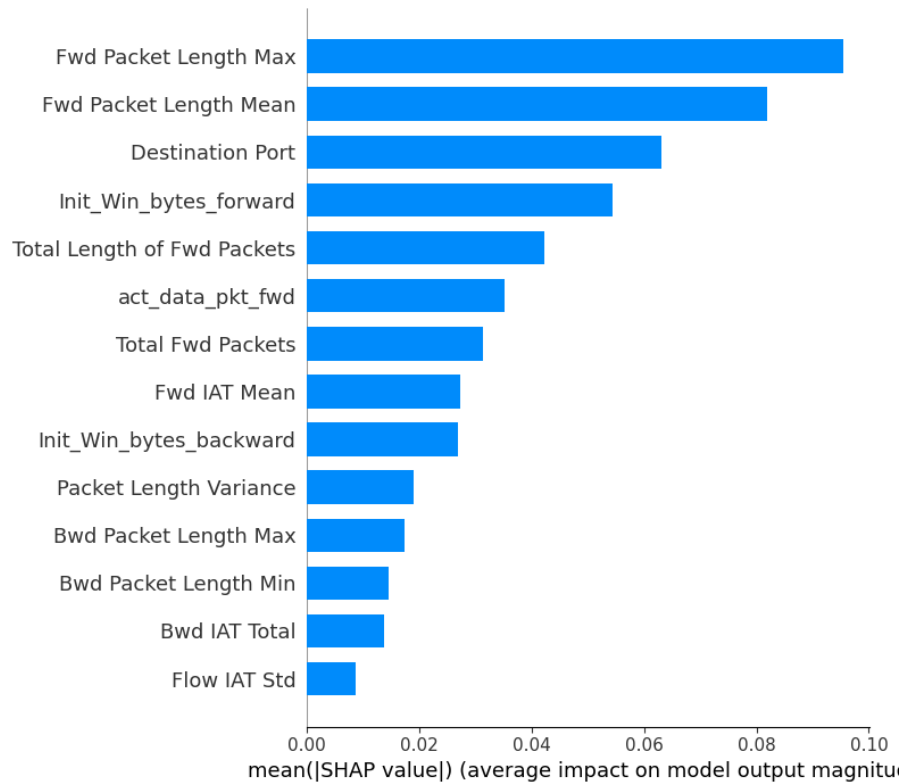


Figure B4: Final 14 features - RF importance and MI scores

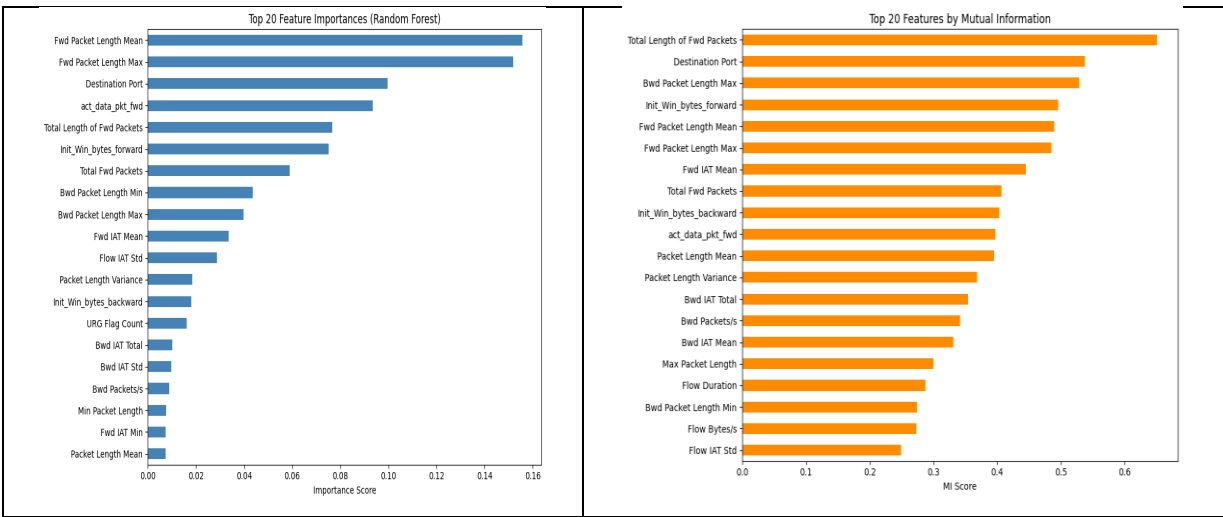


Figure B5: RF feature importance (blue) and Mutual Information score (orange)

Appendix C: Model Training Screenshots

```

Supervised Model: Random Forest

#Model Training - Random Forest Classifier (Production Model)
#Trained on the final 14-feature set (RF importance n MI top-30).

#RF selected for: ensemble variance reduction; majority-vote imbalance
#tolerance; and native SHAP TreeExplainer compatibility (exact Shapley
#values vs. LIME's surrogate approximation).
#n_estimators=100: diminishing returns observed beyond ~200 trees.
#max_depth=None: fully grown trees maximise recall on the attack class;
# ensemble averaging prevents individual tree overfitting.
#random_state=42: fixes bootstrap sampling for reproducibility.
#n_jobs=-1: parallelises tree building across all available CPU cores.
print("\nTraining Random Forest classifier...")
clf = RandomForestClassifier(
    n_estimators=100,
    max_depth=None,
    random_state=42,
    n_jobs=-1
)
clf.fit(X_train_final, y_train)

#Hard class predictions for evaluation; test set was never seen during
#feature selection, preserving evaluation integrity.
y_pred_rf = clf.predict(X_test_final)

#Soft probability scores for DDoS class - required for ROC-AUC computation
#and to allow decision-threshold tuning post-deployment if needed.
y_prob_rf = clf.predict_proba(X_test_final)[: , 1]

#Full per-class report: precision and recall on DDoS are the primary metrics -
#a false negative (missed attack) is operationally costlier than a false positive.
print("\n-- Random Forest Classification Report --")
print(classification_report(y_test, y_pred_rf, target_names=['BENIGN', 'DDoS']))

#Confusion matrix heatmap: raw counts reveal the exact split of
#TN / FP / FN / TP, which Table 1 in the report is drawn from.
cm_rf = confusion_matrix(y_test, y_pred_rf)
plt.figure(figsize=(6, 4))
sns.heatmap(cm_rf, annot=True, fmt='d', cmap='Blues',
            xticklabels=['BENIGN', 'DDoS'], yticklabels=['BENIGN', 'DDoS'])
plt.title('Confusion Matrix - Random Forest')
plt.ylabel('Actual'); plt.xlabel('Predicted')
plt.tight_layout(); plt.show()

✓ 3.5s Python

```

Figure C1: Random Forest training code - 100 trees, *max_depth=None*

```

#Model Training - Isolation Forest (Zero-Day / Unsupervised Layer)
#Label-free anomaly detection addressing CISO requirement iv.

#Normalise features before IF training - without scaling, high-range features
#dominate random partitioning and low-range features are effectively ignored.
#fit_transform() on train; transform()-only on test prevents leakage.
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train_final)
X_test_scaled = scaler.transform(X_test_final)

#Fallback if contamination_ratio was not computed in an earlier cell.
if 'contamination_ratio' not in dir():
    contamination_ratio = 0.45

#Cap at 0.45 - scikit-learn requires contamination strictly < 0.5.
#Training ratio (0.5671) exceeds this limit so capping is necessary.
contamination_ratio = float(min(contamination_ratio, 0.45))
assert 0 < contamination_ratio < 0.5, "contamination_ratio out of valid range"

#n_estimators=100: consistent with RF for fair comparison.
#contamination: derived from training labels only - not hardcoded.
#random_state=42: reproducibility; n_jobs=-1: full CPU parallelisation.
print(f"\nTraining Isolation Forest (contamination={contamination_ratio:.4f})...")
iso = IsolationForest(
    n_estimators=100,
    contamination=contamination_ratio,
    random_state=42,
    n_jobs=-1
)
iso.fit(X_train_scaled)

#IF outputs -1 (anomaly) or +1 (normal); remap to 1=DDoS, 0=BENIGN
#to align with the binary label convention used throughout the pipeline.
y_pred_if_raw = iso.predict(X_test_scaled)
y_pred_if = np.where(y_pred_if_raw == -1, 1, 0)

```

Figure C2: Isolation Forest training code - part 1

```

print("\n--- Isolation Forest Classification Report ---")
print(classification_report(y_test, y_pred_if, target_names=['BENIGN', 'DDoS']))

#Confusion matrix - Oranges palette distinguishes IF from RF (Blues) visually.
cm_if = confusion_matrix(y_test, y_pred_if)
plt.figure(figsize=(6, 4))
sns.heatmap(cm_if, annot=True, fmt='d', cmap='Oranges',
            xticklabels=['BENIGN', 'DDoS'], yticklabels=['BENIGN', 'DDoS'])
plt.title('Confusion Matrix - Isolation Forest')
plt.ylabel('Actual'); plt.xlabel('Predicted')
plt.tight_layout(); plt.show()

#Negate decision_function scores so that higher = more anomalous,
#matching the convention expected by roc_curve (higher score = positive class).
scores_if = -iso.decision_function(X_test_scaled)
fpr_if, tpr_if, _ = roc_curve(y_test, scores_if)
roc_auc_if = auc(fpr_if, tpr_if)

#AUC < 0.5 is expected here: IF treats the majority class (DDoS) as anomalous,
#inverting the score ordering relative to the true label convention.
plt.figure(figsize=(6, 4))
plt.plot(fpr_if, tpr_if, color='darkorange', lw=2, label=f'ROC AUC = {roc_auc_if:.4f}')
plt.plot([0, 1], [0, 1], 'k--')
plt.xlabel('False Positive Rate'); plt.ylabel('True Positive Rate')
plt.title('ROC Curve - Isolation Forest'); plt.legend(loc='lower right')
plt.tight_layout(); plt.show()

print(f"IF ROC-AUC: {roc_auc_if:.4f}")

```

Figure C3: Isolation Forest training code - part 2, contamination=0.45

Appendix D: SHAP Code Screenshots

SHAP Example 1: True DDoS correctly flagged

```
#Explainability – SHAP Waterfall Plot (True Positive Example)
#Individual prediction explanation for Appendix J, Figure J1.
#Demonstrates how SHAP surfaces the exact features driving a confirmed
#DDoS alert – directly supporting SOC analyst trust and alert triage.

#Convert to NumPy arrays for index-based boolean masking – pandas Series
#index alignment can cause silent mismatches when using np.where().
y_test_arr = np.array(y_test)
y_pred_arr = np.array(y_pred_rf)

#Locate the first test sample that is a genuine DDoS flow correctly
#predicted as DDoS (True Positive) – used as Example 1 in the report.
tp_indices = np.where((y_test_arr == 1) & (y_pred_arr == 1))[0]
tp_sample = X_test_final.iloc[[tp_indices[0]]]

#Compute a full SHAP Explanation object for this single sample.
#explainer() (callable syntax) returns richer metadata than
#explainer.shap_values(), enabling waterfall_plot() rendering.
shap_explanation = explainer(tp_sample)

#Handle both SHAP API versions – same convention as Cell 7.
#3D array (new API): slice last axis to isolate DDoS class (index 1).
#2D array (old API): already class-specific, no slicing needed.
if shap_explanation.values.ndim == 3:
    shap_exp_class1 = shap_explanation[:, :, 1] # new API
else:
    shap_exp_class1 = shap_explanation # old API

#Waterfall plot decomposes this single prediction into per-feature
#contributions – red bars push toward DDoS, blue bars push toward BENIGN.
#The base value is the model's mean prediction across the training set.
print("\nExample 1: True DDoS attack correctly flagged")
shap.waterfall_plot(shap_exp_class1[0])
```

✓ 0.4s Python

Figure D1: SHAP global feature importance code - TreeExplainer, beeswarm plot

SHAP Example 2: False positive or second TP

```
#CELL 9: Explainability – SHAP Waterfall Plot (False Positive Example)
#Individual prediction explanation for Appendix J, Figure J2.
#Shows which features drove a BENIGN flow to be misclassified as DDoS –
#the key output for demonstrating alert fatigue reduction to the CISO.

#Locate test samples where the true label is BENIGN (0) but the RF
#predicted DDoS (1) – these are false positives (FP).
#With RF's near-perfect precision, FPs are rare (2 in 45,149 samples),
#so the existence check is necessary before attempting to index.
fp_indices = np.where((y_test_arr == 0) & (y_pred_arr == 1))[0]

if len(fp_indices) > 0:
    #Take the first FP sample – used as Example 2 in the report.
    fp_sample = X_test_final.iloc[[fp_indices[0]]]
    fp_explanation = explainer(fp_sample)

    #Same API compatibility handling as Cells 7 and 8.
    if fp_explanation.values.ndim == 3:
        fp_exp_class1 = fp_explanation[:, :, 1] # new API
    else:
        fp_exp_class1 = fp_explanation # old API

    #Waterfall plot reveals the competing SHAP forces behind the misclassification –
    #the dominant negative driver (pushing toward BENIGN) is the key analyst insight
    #that explains why the alert fired incorrectly.
    print("\nExample 2: BENIGN traffic incorrectly flagged as DDoS (false positive)")
    shap.waterfall_plot(fp_exp_class1[0])

else:
    #Fallback: if no FPs exist in this split (possible with near-perfect RF),
    #show a second TP so the waterfall plot section of the report is still
    #populated with a meaningful individual prediction example.
    print("\nNo false positives in this test sample. Showing a second TP for comparison.")
    tp_sample2 = X_test_final.iloc[[tp_indices[1]]]
    tp2_explanation = explainer(tp_sample2)

    #Same API compatibility handling as above.
    if tp2_explanation.values.ndim == 3:
        tp2_exp_class1 = tp2_explanation[:, :, 1] # new API
    else:
        tp2_exp_class1 = tp2_explanation # old API

    shap.waterfall_plot(tp2_exp_class1[0])
```

✓ 0.2s

Python

Figure D2: SHAP individual prediction code - Example 1 (TP) and Example 2 (FP) waterfall plots

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